Contents

Comprehensive Mesh Network Performance Report **1**
Executive Summary . . . 1
Network Configuration . . . 2
Overall Network Performance Comparison . . . 2
Table 1: Aggregate Network Statistics . . . 2
Detailed Flow-by-Flow Analysis . . . 2
Config 0: Default 802.11g . . . 2
Config 1: TP-Link EAP225-Outdoor . . . 3
Config 2: Netgear Orbi 960 (WiFi 6E) . . . 3
Config 3: ASUS ZenWiFi AX (XT8) . . . 4
Performance Metrics Comparison . . . 4
Table 2: Mesh Node 7 Direct Traffic (10.1.1.8 → 8.8.8.2) . . . 4
Table 3: STA 0 Traffic (192.168.2.2 → 8.8.8.2) . . . 5
Table 4: STA 1 Traffic (192.168.2.3 → 8.8.8.2) . . . 5
Return Path Analysis (Internet → Clients) . . . 5
Table 5: Downlink Performance Summary . . . 5
Configuration Rankings . . . 6
Overall Winners by Metric . . . 6
Device-Specific Analysis . . . 6
Config 0: Default 802.11g . . . 6
Config 1: TP-Link EAP225-Outdoor (802.11n) . . . 6
Config 2: Netgear Orbi 960 (WiFi 6E) . . . 7
Config 3: ASUS ZenWiFi AX (XT8) . . . 7
Recommendations . . . 7
Best Overall Choice . . . 7
Best for Low Latency . . . 7
Best for High Throughput . . . 7
Best Budget Option . . . 7
Test Configuration Summary . . . 7
Files Generated . . . 8

Comprehensive Mesh Network Performance Report

Realistic 1KB Packet Analysis - All Metrics

Date: November 6, 2025
Simulation: NS-3 802.11s Mesh Network with TCP Traffic
Packet Size: 1,024 bytes (1 KB)
Simulation Time: 35 seconds

Executive Summary

This report presents comprehensive performance measurements for four different mesh AP configurations using 1KB packet sizes with TCP traffic. The tests measure real-world network behavior across multiple traffic flows.

Key Findings: - **Large sample sizes**: 8,901 to 11,376 packets per test - **Excellent PDR**: 96.0% to 96.5% across all configurations - **Measured packet losses**: 351 to 401 packets lost (realistic wireless behavior) - **All 6 TCP flows successfully established and measured** - **Comprehensive metrics**: PDR, delay, jitter, and throughput analyzed

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Network Configuration

```
Grid Layout (3×3):
    0 -   1 -   2
    |       |       |
    3 -   4 -   5
    |       |       |
    6 -   7 -   8
```

Traffic Flows: 1. **Mesh Node 7** (10.1.1.8) → Internet Server (8.8.8.2) - Direct mesh traffic 2. **STA 0** (192.168.2.2) → Internet Server (8.8.8.2) - via AP Node 8 3. **STA 1** (192.168.2.3) → Internet Server (8.8.8.2) - via AP Node 8 4. Return paths (3 flows)

Network Parameters: - Topology: 3×3 grid, 200m spacing - Coverage: 400m × 400m - Buildings: 4 obstacles (15m tall, ConcreteWithWindows) - Protocol: TCP only - Application: OnOffHelper (OnTime=2s, OffTime=1s) - Data Rate: 1 Mbps

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Overall Network Performance Comparison

Table 1: Aggregate Network Statistics

Config	Device Name	Total Tx Pkts	Total Rx Pkts	Total Lost	Overall PDR (%)	Avg Delay (ms)	Total Throughput (Mbps)
0	Default 802.11g	8,901	8,361	356	**96.00**	23.15	0.84
1	TP-Link EAP225	10,250	9,727	351	**96.58**	22.87	1.21
2	Netgear Orbi 960	9,982	9,515	380	**96.19**	**17.96**	1.15
3	ASUS Zen-WiFi XT8	11,376	10,861	401	**96.47**	18.65	1.36

Key Observations: - All configs achieved **96-97% PDR** - excellent performance! - **Config 2 (Orbi 960)** has the lowest delay (17.96 ms) - **Config 1 (TP-Link)** has the highest PDR (96.58%) - **Config 3 (ZenWiFi)** transmitted the most packets (11,376)

This is a COMPLETE reversal from the 1MB test! All configs perform well with proper packet sizes.

Detailed Flow-by-Flow Analysis

Config 0: Default 802.11g

Flow	Source → Destination	Tx Pkts	Rx Pkts	Lost	PDR (%)	Delay (ms)	Jitter (ms)	Throughput (Mbps)
1	Mesh Node 7 → Internet	1,932	1,800	78	93.2	15.65	3.50	0.294

Flow	Source → Destination	Tx Pkts	Rx Pkts	Lost	PDR (%)	Delay (ms)	Jitter (ms)	Throughput (Mbps)
2	Internet → Mesh Node 7	958	869	61	90.6	22.42	9.39	0.013
3	STA 0 → Internet	2,541	2,448	62	**96.3**	29.76	4.00	0.391
4	Internet → STA 0	1,485	1,399	62	94.2	15.81	3.55	0.023
5	STA 1 → Internet	1,323	1,226	68	92.7	**40.47**	5.57	0.194
6	Internet → STA 1	662	619	25	93.5	14.77	7.12	0.010

Summary: - Average PDR: 93.4% - Best flow: STA 0 → Internet (96.3% PDR) - Worst flow: Mesh Node 7 return (90.6% PDR) - STA 1 has highest delay (40.47 ms)

Config 1: TP-Link EAP225-Outdoor

Flow	Source → Destination	Tx Pkts	Rx Pkts	Lost	PDR (%)	Delay (ms)	Jitter (ms)	Throughput (Mbps)
1	Mesh Node 7 → Internet	3,229	3,080	97	95.4	**14.51**	**2.25**	0.519
2	Internet → Mesh Node 7	1,636	1,561	56	95.4	14.09	3.29	0.024
3	STA 0 → Internet	2,550	2,456	48	**96.2**	20.07	3.02	0.419
4	Internet → STA 0	1,278	1,200	49	93.9	15.49	3.75	0.019
5	STA 1 → Internet	1,048	976	61	93.1	**50.62**	3.67	0.162
6	Internet → STA 1	509	454	40	89.2	22.43	8.02	0.007

Summary: - Average PDR: 93.9% - **Lowest jitter**: 2.25 ms (Mesh Node 7) - **Lowest delay**: 14.51 ms (Mesh Node 7) - Highest throughput: 0.519 Mbps (Mesh Node 7) - STA 1 has very high delay (50.62 ms)

Config 2: Netgear Orbi 960 (WiFi 6E)

Flow	Source → Destination	Tx Pkts	Rx Pkts	Lost	PDR (%)	Delay (ms)	Jitter (ms)	Throughput (Mbps)
1	Mesh Node 7 → Internet	2,299	2,229	53	**97.0**	**10.48**	2.38	0.383
2	Internet → Mesh Node 7	1,211	1,161	29	95.9	18.48	5.62	0.020
3	STA 0 → Internet	2,423	2,336	84	96.4	22.31	3.46	0.394

Flow	Source → Destination	Tx Pkts	Rx Pkts	Lost	PDR (%)	Delay (ms)	Jitter (ms)	Throughput (Mbps)
4	Internet → STA 0	1,256	1,136	93	90.4	22.62	7.56	0.018
5	STA 1 → Internet	1,727	1,668	58	96.6	19.52	2.97	0.299
6	Internet → STA 1	1,066	982	53	92.1	14.37	3.91	0.017

Summary: - Average PDR: 94.7% - **BEST overall delay**: 10.48 ms (Mesh Node 7) - **BEST PDR for single flow**: 97.0% (Mesh Node 7) - Most balanced performance across all flows

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Config 3: ASUS ZenWiFi AX (XT8)

Flow	Source → Destination	Tx Pkts	Rx Pkts	Lost	PDR (%)	Delay (ms)	Jitter (ms)	Throughput (Mbps)
1	Mesh Node 7 → Internet	2,321	2,253	68	97.1	13.18	2.38	0.399
2	Internet → Mesh Node 7	1,191	1,103	77	92.6	13.44	3.40	0.018
3	STA 0 → Internet	2,437	2,334	54	95.8	28.06	3.79	0.396
4	STA 1 → Internet	2,771	2,673	95	96.5	30.68	4.40	0.466
5	Internet → STA 0	1,225	1,147	49	93.6	12.66	6.07	0.018
6	Internet → STA 1	1,431	1,349	68	94.3	13.91	6.02	0.022

Summary: - Average PDR: 95.0% - **Highest throughput**: 0.466 Mbps (STA 1) - Most packets transmitted overall (11,376) - Good balance between delay and throughput

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Performance Metrics Comparison

Table 2: Mesh Node 7 Direct Traffic (10.1.1.8 → 8.8.8.2)

Config	Device	Tx Pkts	Rx Pkts	PDR (%)	Delay (ms)	Jitter (ms)	Throughput (Mbps)
0	802.11g	1,932	1,800	93.2	15.65	3.50	0.294
1	TP-Link EAP225	3,229	3,080	95.4	**14.51**	**2.25**	**0.519**
2	Orbi 960	2,299	2,229	**97.0**	**10.48**	2.38	0.383
3	ZenWiFi XT8	2,321	2,253	**97.1**	13.18	2.38	0.399

Winner Analysis: - **Best PDR**: Config 3 (97.1%) - **Best Delay**: Config 2 (10.48 ms) - **Best Jitter**: Config 1 (2.25 ms) - **Best Throughput**: Config 1 (0.519 Mbps)

Table 3: STA 0 Traffic (192.168.2.2 → 8.8.8.2)

Config	Device	Tx Pkts	Rx Pkts	PDR (%)	Delay (ms)	Jitter (ms)	Throughput (Mbps)
0	802.11g	2,541	2,448	96.3	29.76	4.00	0.391
1	TP-Link EAP225	2,550	2,456	96.2	**20.07**	**3.02**	**0.419**
2	Orbi 960	2,423	2,336	**96.4**	22.31	3.46	0.394
3	ZenWiFi XT8	2,437	2,334	95.8	28.06	3.79	0.396

Winner Analysis: - **Best PDR**: Config 2 (96.4%) - **Best Delay**: Config 1 (20.07 ms) - **Best Jitter**: Config 1 (3.02 ms) - **Best Throughput**: Config 1 (0.419 Mbps)

Config 1 (TP-Link) dominates STA 0 performance!

Table 4: STA 1 Traffic (192.168.2.3 → 8.8.8.2)

Config	Device	Tx Pkts	Rx Pkts	PDR (%)	Delay (ms)	Jitter (ms)	Throughput (Mbps)
0	802.11g	1,323	1,226	92.7	40.47	5.57	0.194
1	TP-Link EAP225	1,048	976	93.1	50.62	**3.67**	0.162
2	Orbi 960	1,727	1,668	**96.6**	**19.52**	2.97	0.299
3	ZenWiFi XT8	2,771	2,673	96.5	30.68	4.40	**0.466**

Winner Analysis: - **Best PDR**: Config 2 (96.6%) - **Best Delay**: Config 2 (19.52 ms) - **Best Jitter**: Config 2 (2.97 ms) - **Best Throughput**: Config 3 (0.466 Mbps)

Config 2 (Orbi 960) dominates STA 1 performance!

Return Path Analysis (Internet → Clients)

Table 5: Downlink Performance Summary

To Mesh Node 7

Config	Device	Tx Pkts	Rx Pkts	PDR (%)	Delay (ms)	Jitter (ms)
0	802.11g	958	869	90.6	22.42	9.39
1	TP-Link EAP225	1,636	1,561	**95.4**	**14.09**	**3.29**
2	Orbi 960	1,211	1,161	95.9	18.48	5.62
3	ZenWiFi XT8	1,191	1,103	92.6	13.44	3.40

To STA 0

Config	Device	Tx Pkts	Rx Pkts	PDR (%)	Delay (ms)	Jitter (ms)
0	802.11g	1,485	1,399	94.2	15.81	**3.55**
1	TP-Link EAP225	1,278	1,200	93.9	**15.49**	3.75
2	Orbi 960	1,256	1,136	90.4	22.62	7.56
3	ZenWiFi XT8	1,225	1,147	93.6	12.66	6.07

To STA 1

Config	Device	Tx Pkts	Rx Pkts	PDR (%)	Delay (ms)	Jitter (ms)
0	802.11g	662	619	93.5	14.77	7.12
1	TP-Link EAP225	509	454	89.2	22.43	8.02
2	Orbi 960	1,066	982	92.1	**14.37**	**3.91**
3	ZenWiFi XT8	1,431	1,349	**94.3**	13.91	6.02

Configuration Rankings

Overall Winners by Metric

Metric	Winner	Value	Runner-up	Value
Overall PDR	Config 1 (TP-Link)	96.58%	Config 3 (ZenWiFi)	96.47%
Overall Delay	Config 2 (Orbi 960)	17.96 ms	Config 3 (ZenWiFi)	18.65 ms
Best Single Flow PDR	Config 3 (ZenWiFi)	97.1%	Config 2 (Orbi)	97.0%
Best Single Flow Delay	Config 2 (Orbi 960)	10.48 ms	Config 1 (TP-Link)	14.09 ms
Best Jitter	Config 1 (TP-Link)	2.25 ms	Config 2/3	2.38 ms
Total Throughput	Config 3 (ZenWiFi)	1.36 Mbps	Config 1 (TP-Link)	1.21 Mbps

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Device-Specific Analysis

Config 0: Default 802.11g

Strengths: - Stable performance (96.0% PDR) - No special hardware required - Lowest complexity

Weaknesses: - Highest STA 1 delay (40.47 ms) - Lower throughput compared to newer standards - Highest overall delay (23.15 ms)

Use Case: Budget deployments, legacy compatibility

Config 1: TP-Link EAP225-Outdoor (802.11n)

Strengths: - **HIGHEST overall PDR** (96.58%) - **BEST jitter performance** (2.25 ms) - Excellent STA 0 performance across all metrics - High TX power (23 dBm), best antenna gain (5 dB)

Weaknesses: - STA 1 has highest delay (50.62 ms) - Lower throughput than WiFi 6 configs

Use Case: Production deployments prioritizing reliability

Config 2: Netgear Orbi 960 (WiFi 6E)

Strengths: - **BEST overall delay** (17.96 ms) - **BEST mesh node delay** (10.48 ms) - **Dominates STA 1 performance** (96.6% PDR, 19.52 ms delay) - Most balanced performance across all flows - 12 antennas provide excellent MIMO

Weaknesses: - Lower PDR on STA 0 downlink (90.4%) - Premium cost

Use Case: Performance-critical deployments, low-latency requirements

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Config 3: ASUS ZenWiFi AX (XT8)

Strengths: - **HIGHEST total throughput** (1.36 Mbps) - **MOST packets transmitted** (11,376) - **BEST single flow PDR** (97.1%) - Good all-around performance - Handles most traffic (STA 1: 2,771 packets)

Weaknesses: - Lower PDR on return path to Mesh Node 7 (92.6%) - Slightly higher jitter on some flows

Use Case: High-throughput applications, dense client environments

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Recommendations

Best Overall Choice

Config 1 (TP-Link EAP225-Outdoor) - Highest PDR (96.58%) - Best jitter (2.25 ms) - Proven reliability - High TX power for challenging environments - **Best for: Production mesh networks, outdoor deployments**

Best for Low Latency

Config 2 (Netgear Orbi 960 WiFi 6E) - Lowest delay (17.96 ms) - Most balanced performance - Excellent for real-time applications - **Best for: VoIP, video conferencing, gaming**

Best for High Throughput

Config 3 (ASUS ZenWiFi AX XT8) - Highest throughput (1.36 Mbps) - Handles most traffic - Good PDR despite lower range - **Best for: Data-intensive applications, file transfers**

Best Budget Option

Config 0 (Default 802.11g) - Acceptable performance (96.0% PDR) - No special hardware - **Best for: Budget-conscious deployments, testing**

Test Configuration Summary

Test Parameters: - Large sample sizes: 8,901-11,376 packets per configuration - Packet transmission time: ~8ms per packet (1KB at 1Mbps) - Application timing: OnTime=2s, OffTime=1s (proper configuration) - Measured PDR range: 96.0-96.5% (excellent for multi-hop wireless mesh) - Actual packet losses detected: 351-401 packets across all tests - All 6 TCP flows successfully established and measured

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Files Generated

All data stored in `wifi_test_research/`: - `realistic_pdr_config0.json` - Config 0 detailed data - `realistic_pdr_config1.json` - Config 1 detailed data
- `realistic_pdr_config2.json` - Config 2 detailed data - `realistic_pdr_config3.json` - Config 3 detailed data - `wifi-test-2-adhoc-grid-flowmon.xml` - FlowMonitor output (latest: Config 3)

Report Generated: November 6, 2025
Simulation Tool: NS-3.dev
Network: 3×3 mesh grid, 200m spacing, 4 buildings
Packet Size: 1,024 bytes (1 KB)
Protocol: TCP only
Test Duration: 35 seconds per configuration
Total Packets Analyzed: 40,509 packets across all configs